FEASIFY CAPSTONE PANEL BRIEFING & DEFENSE SUMMARY

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Audience: Academic Defense Panel, Technical Evaluators, Capstone Advisers

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1. Project At A Glance

- **Project Name:** Feasify (Web-Based Capstone Proposal & Feasibility Management System)
- **Author / Team:** Feasify Capstone Development Team
- **Objective:** Streamline the submission, financial calculation, AI evaluation, and faculty review of student feasibility studies.
- **Core Technology Stack:** React 19, TypeScript, Tailwind CSS 4, Vite 7, Node.js v22, Express 5, Firebase Cloud Platform, Leaflet OpenStreetMap, and Google Gemini AI.

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2. Key Testing Findings & Empirical Metrics

Quality Metric	Empirical Finding
Total Modules Audited	14 Modules
Total Test Scenarios Executed	67 Scenarios (42 Functional + 25 Auto)
Total Automated Assertions Passed	105 Assertions
Test Pass Rate	89.6% (60 Passed / 67 Executed)
Production Build Status	PASSED (Vite built client in 50.32s)
Local Endpoint Latency (Baseline)	4.0 milliseconds
Concurrency Target Tested	50 Simultaneous Virtual Users
Overall Capstone Readiness Verdict	CONDITIONALLY READY FOR DEMONSTRATION

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3. Verified System Strengths

1. **Strict Maximum 3-Proposal Rule Enforced in UI & Backend:**
   - Groups cannot submit a 4th proposal. The system locks creation upon reaching three proposals and accurately retains all existing records upon page reload.
2. **Mathematically Rigorous Financial Projections:**
   - The fundamental balance sheet accounting equation ($\text{Assets} = \text{Liabilities} + \text{Equity}$) holds across all five projected operational years with zero mathematical discrepancies.

3. **Philippine Commercial Trademark Protection:**
   - Automated verification flags student queries against famous Philippine brand trademarks (e.g., Goldilocks, Cocopan, Mega Sardines, Red Ribbon, 7-Eleven) to prevent intellectual property violations.
4. **Zero Commercial API Billing Dependencies for Maps:**
   - Uses Leaflet and OpenStreetMap tiles with embedded Philippine landmarks, eliminating Google Cloud billing overhead and quota failure risks.

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4. Key Limitations Disclosed for Defense Transparency

1. **Security Vulnerability: Plaintext Passwords in Firestore (`SEC-01`):**
   - *Status:* Identified in `ChairpersonModule.tsx` during Excel roster imports.
   - *Mitigation for Defense:* Safe for demo with mock test accounts; must be removed before production deployment.
2. **Security Vulnerability: Unauthenticated Proposal Approval API (`SEC-02`):**
   - *Status:* Backend `PUT /api/proposals/:id/status` endpoint lacks token validation.
   - *Mitigation for Defense:* In demonstration on a local or closed WiFi network, demonstrate faculty approvals via the legitimate Adviser UI.
3. **Student Username Edit Error (`FEASIFY-BUG-002`):**
   - *Status:* A missing variable declaration in `Settings.tsx` prevents student username updates.
   - *Mitigation for Defense:* Avoid demonstrating student username changes in Settings until the two-line state declaration fix is applied.

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5. Recommendation for Panel Members

The testing team confirms that Feasify is **Conditionally Ready for Capstone Demonstration**. The functional core is robust, well-architected, and fully aligned with academic requirements.
